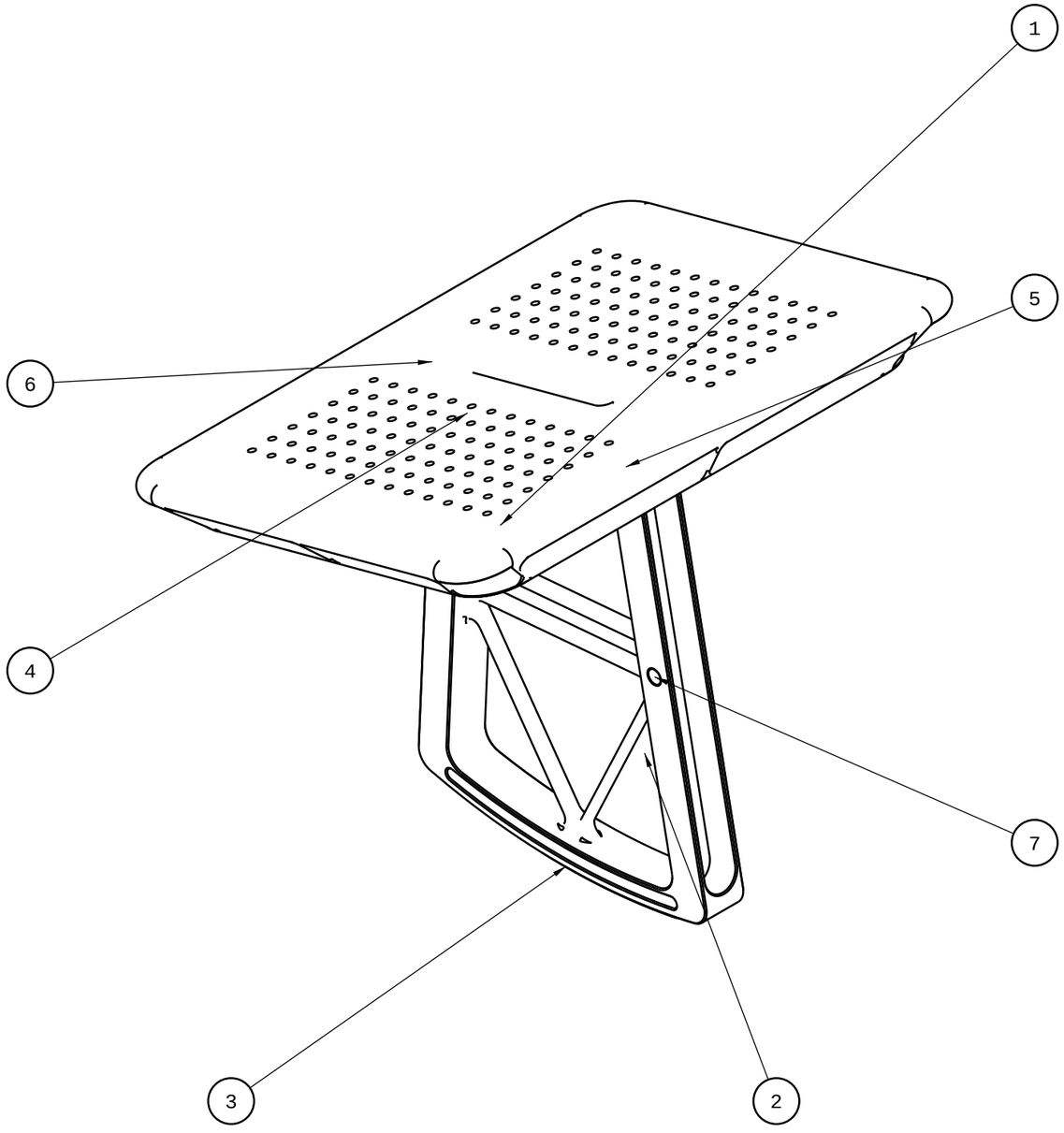
Parts List			
Item	Qty	Part Number	Material
1	1	ULTRA TOP	6061-T6 AL
2	1	ULTRA LEG	6061-T6 AL
3	1	BUMPON (3M, TPR)	TPR
4	2	MAGNET D6 x 6	NdFeB N52, Ni-Cu-Ni
5	2	MAGNET D6 x 3	NdFeB N52, Ni-Cu-Ni
6	2	MAGNET D6 x 1	NdFeB N52, Ni-Cu-Ni
7	4	MAGNET D6 x 1.5	NdFeB N52, Ni-Cu-Ni

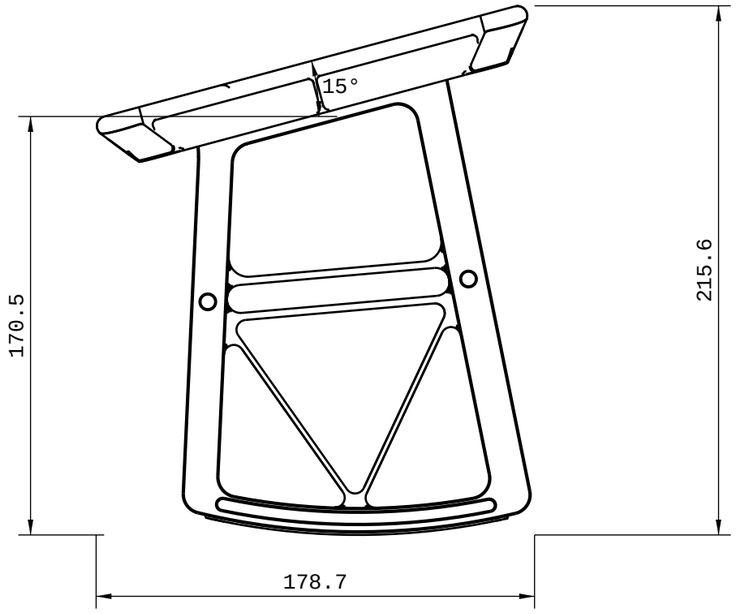


ISOMETRIC VIEW, T (STANDING) CONFIGURATION, SCALE 1:2

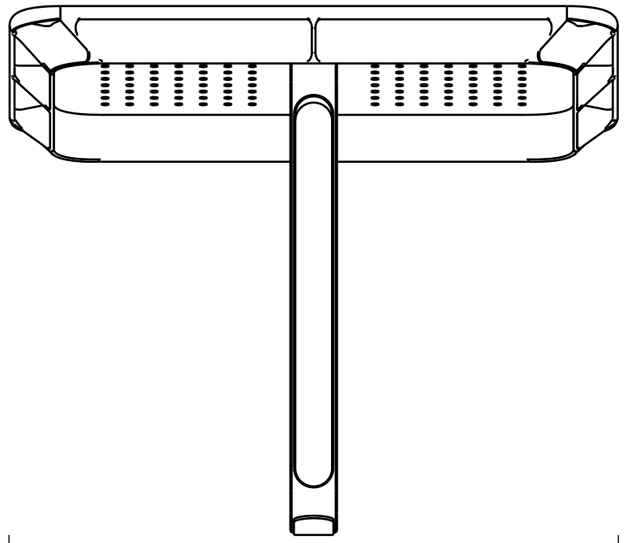
NOTES:

1. MATERIAL: 6061-T6 ALUMINUM. MACHINE FROM SOLID.
2. FINISH: AS MACHINED, DEBURR AND BREAK ALL EDGES 0.2 MAX.
3. UNLESS NOTED: LINEAR +/-0.1, ANGLES +/-0.5 DEG.
4. STEP FILE 'Ultra Bench T.step' IS THE MASTER; THIS DRAWING CALLS OUT OVERALL AND CRITICAL DIMENSIONS ONLY.
5. MAGNET BORES: D6.40 +0.05/-0, DEPTH AS NOTED. MAGNETS SEAT ON BORE FLOOR; BOND IN WITH TWO-PART EPOXY.
6. MAGNET POLARITY: TOP MAGNETS M1-M4 N POLE FACES OUT (TOWARD LEG). LEG MAGNETS M5-M10 S POLE FACES OUT. SEE SHEET 8. VERIFY ATTRACTION BEFORE BONDING.
7. TENON (TOP) / SOCKET (LEG) IS A SLIDING FIT: 10.90 / 11.00.
8. DO NOT SCALE DRAWING. DIMENSIONS IN MM.

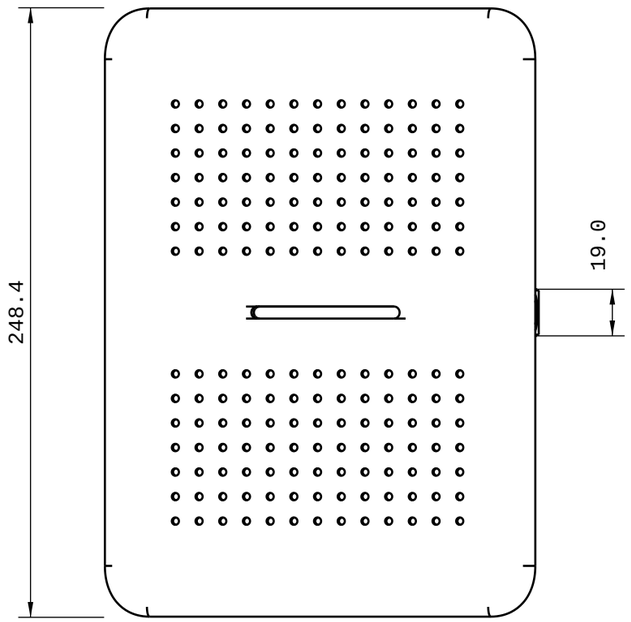
DEPT. CW&T	TECHNICAL REFERENCE	CREATED BY Che-Wei Wang	APPROVED BY 10/16/26
		DOCUMENT TYPE Fabrication drawing - ASSEMBLY	DOCUMENT STATUS FOR FABRICATION
CW&T		TITLE Ultra Bench T	DWG NO. UBT-001
		REV. A	DATE OF ISSUE 10/16/26
			SHEET 1/8



FRONT



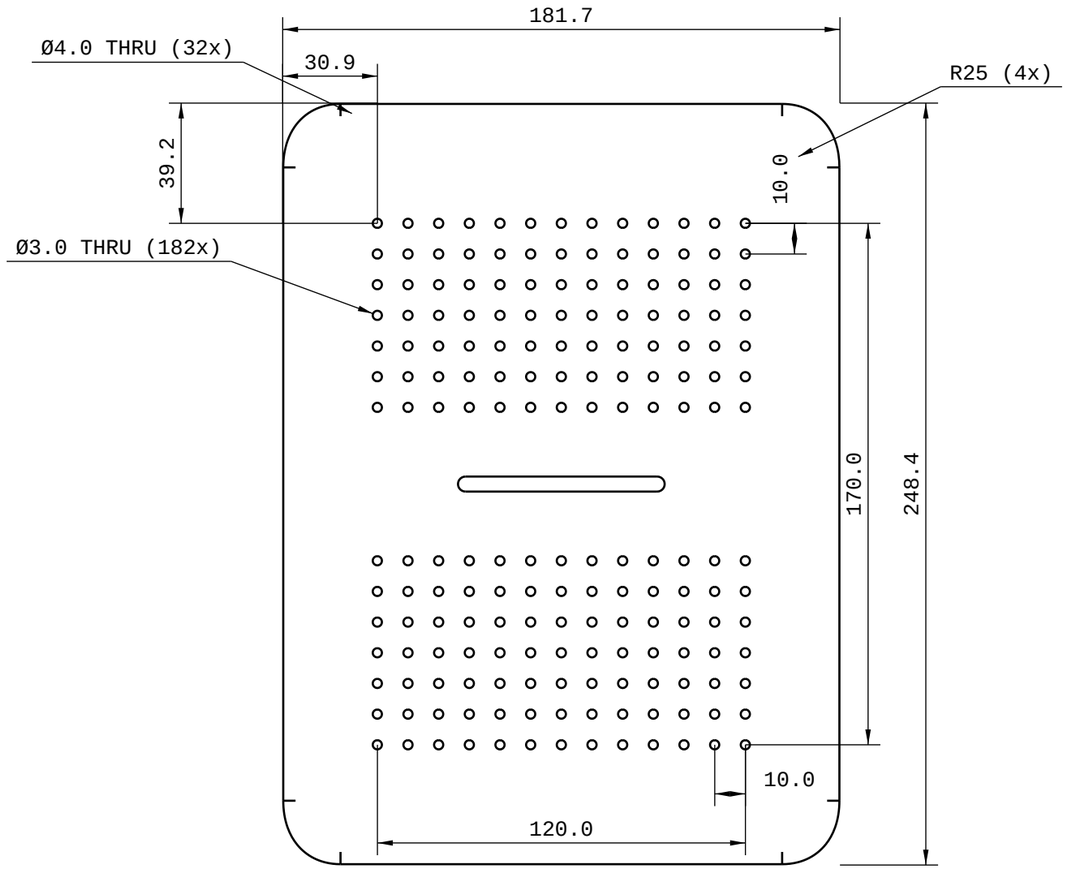
RIGHT



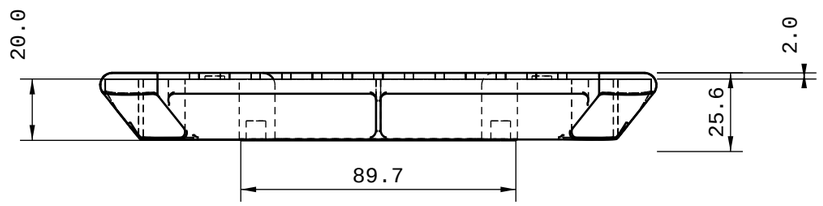
TOP

ASSEMBLY, T CONFIGURATION. SCALE 2:5.
LEG STANDS ON BUMPON; SEAT TILTS 15 DEG.
OVERALL DIMENSIONS FOR REFERENCE;
PART DIMENSIONS ON SHEETS 3-7.

DEPT. CW&T	TECHNICAL REFERENCE	CREATED BY Che-Wei Wang	APPROVED BY 10/16/26
		DOCUMENT TYPE Fabrication drawing - ASSEMBLY OVERALL	DOCUMENT STATUS FOR FABRICATION
CW&T		TITLE Ultra Bench T	DWG NO. UBT-001
		REV. A	DATE OF ISSUE 10/16/26
			SHEET 2/8

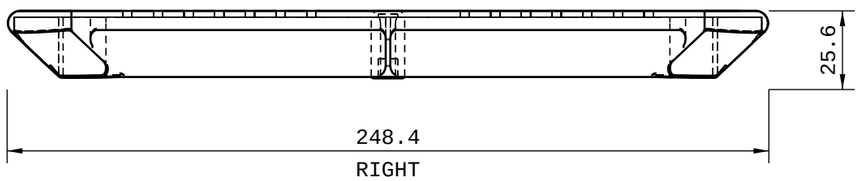


TOP (DECK UP)



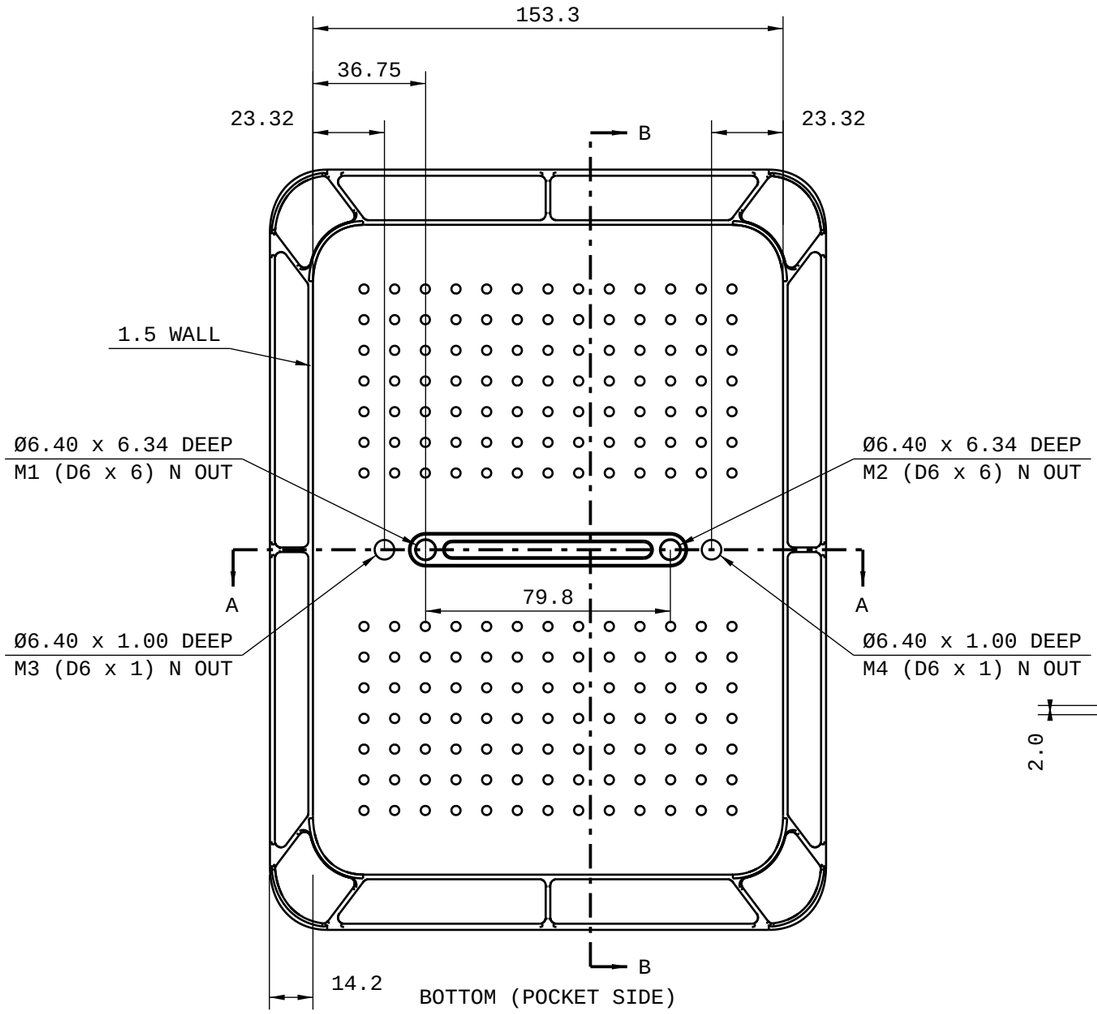
FRONT

ULTRA TOP, SCALE 1:2. DRAWN IN DECK FRAME
(DECK HORIZONTAL; TILT 15 DEG IS SET BY THE LEG).
D3 PERFORATIONS ON 10 x 10 GRID, 13 COLUMNS x 14 ROWS
(7 + 7 ROWS, 50 GAP ON CENTRE LINE).
UNDERSIDE AND SECTIONS: SHEET 4.

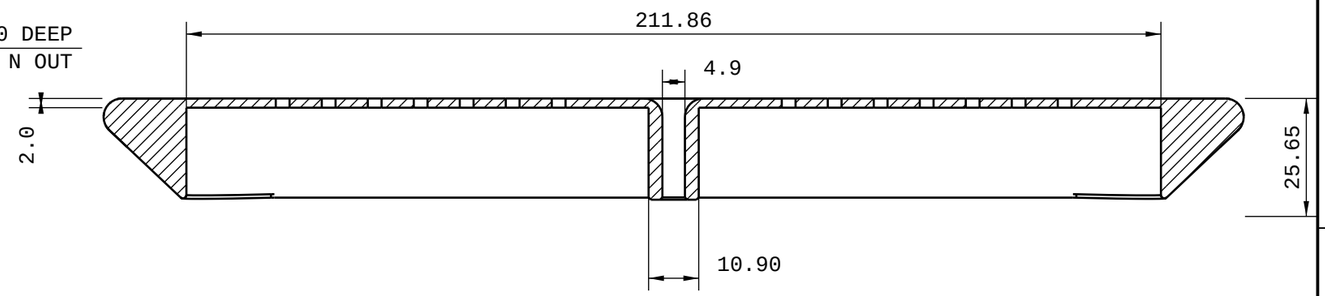


RIGHT

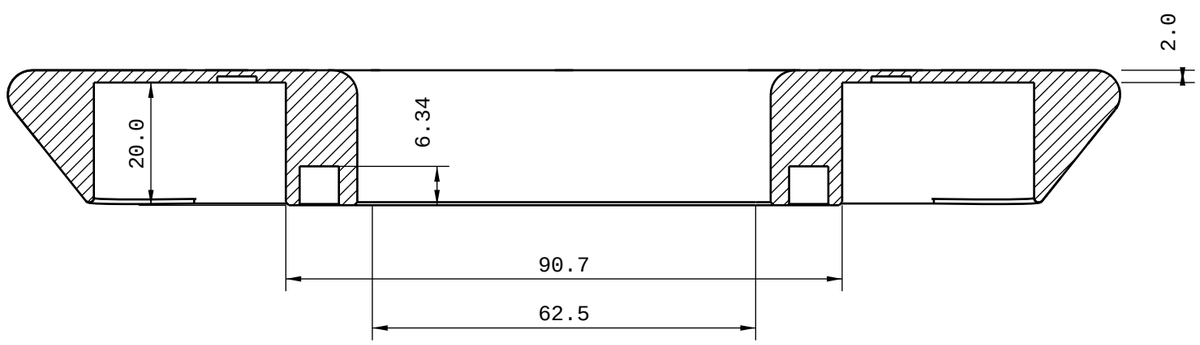
DEPT. CW&T	TECHNICAL REFERENCE	CREATED BY Che-Wei Wang	APPROVED BY 10/16/26
		DOCUMENT TYPE Fabrication drawing - ULTRA TOP	DOCUMENT STATUS FOR FABRICATION
CW&T		TITLE Ultra Bench T	DWG NO. UBT-001
		REV. A	DATE OF ISSUE 10/16/26
			SHEET 3/8



BOTTOM (POCKET SIDE)



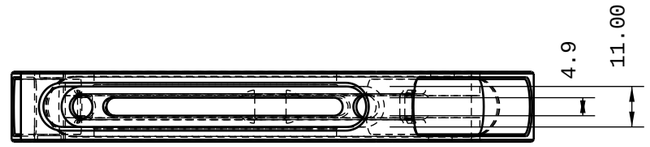
SECTION B-B, SCALE 3:4

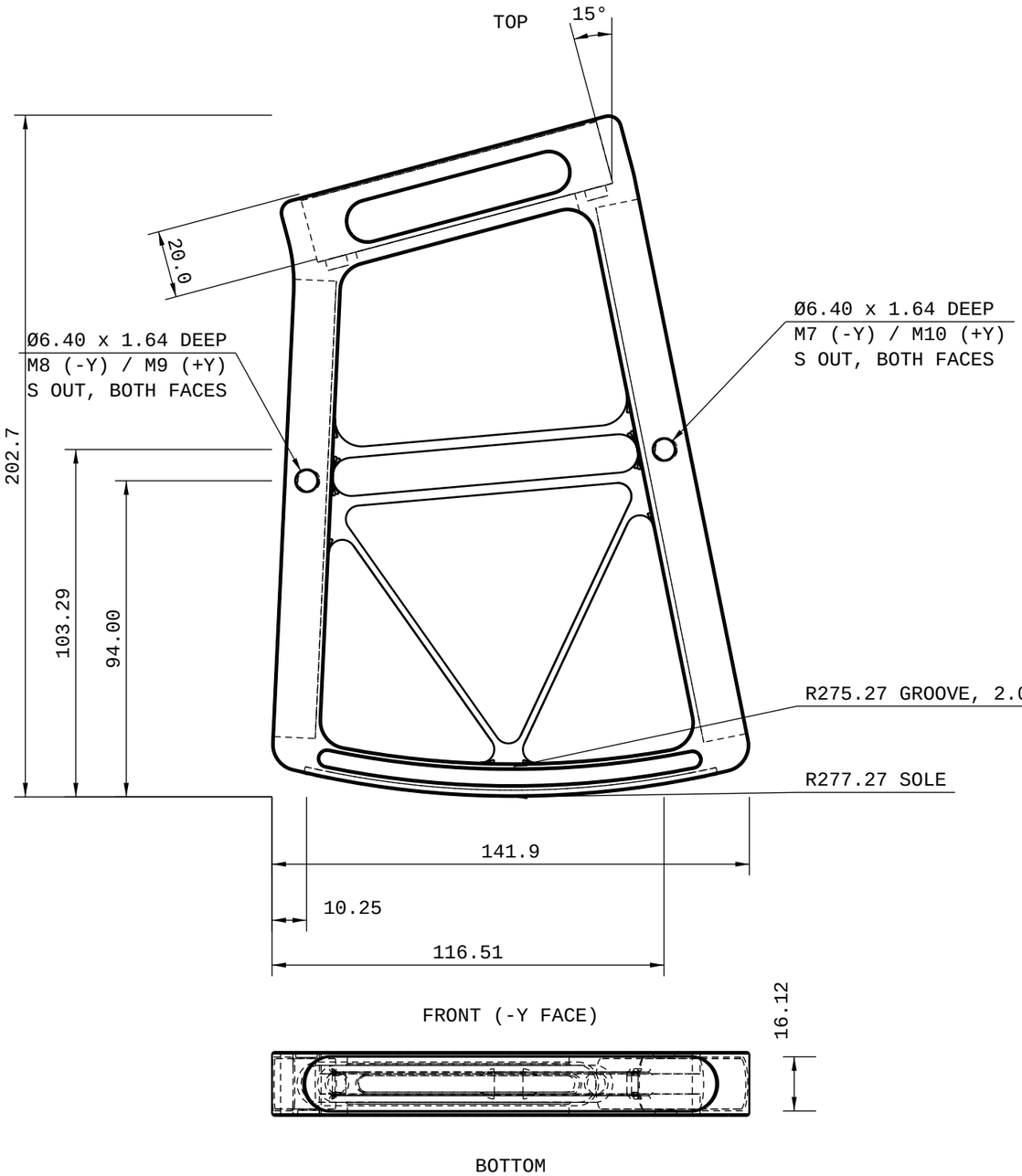


SECTION A-A, SCALE 1:1

ULTRA TOP UNDERSIDE, SCALE 1:2. DECK FRAME.
RIB WALLS 1.5 THK (SEE STEP). MAGNET BORES
D6.40 +0.05/-0 AT DEPTHS NOTED; M1/M2 ARE IN
THE TENON END FACES, M3/M4 IN THE POCKET CEILING.

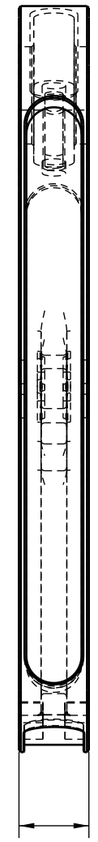
DEPT. CW&T	TECHNICAL REFERENCE	CREATED BY Che-Wei Wang	APPROVED BY 10/16/26
		DOCUMENT TYPE Fabrication drawing - ULTRA TOP UNDERSIDE	DOCUMENT STATUS FOR FABRICATION
CW&T		TITLE Ultra Bench T	DWG NO. UBT-001
		REV. A	DATE OF ISSUE 10/16/26 / SHEET 4/8





FRONT (-Y FACE)

BOTTOM

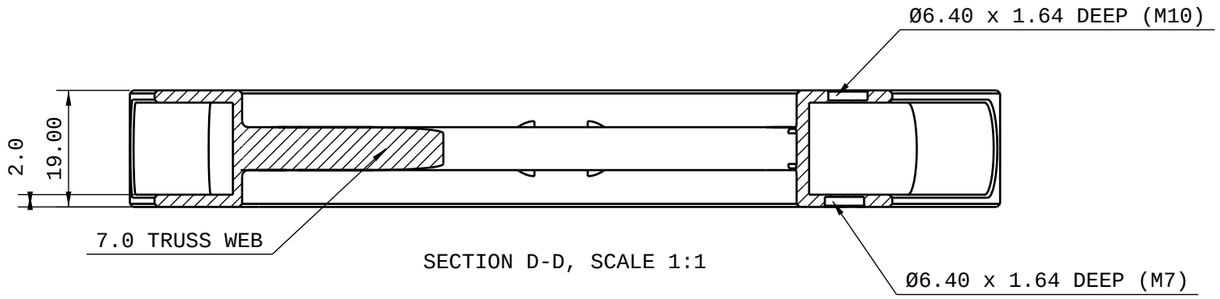


19.00

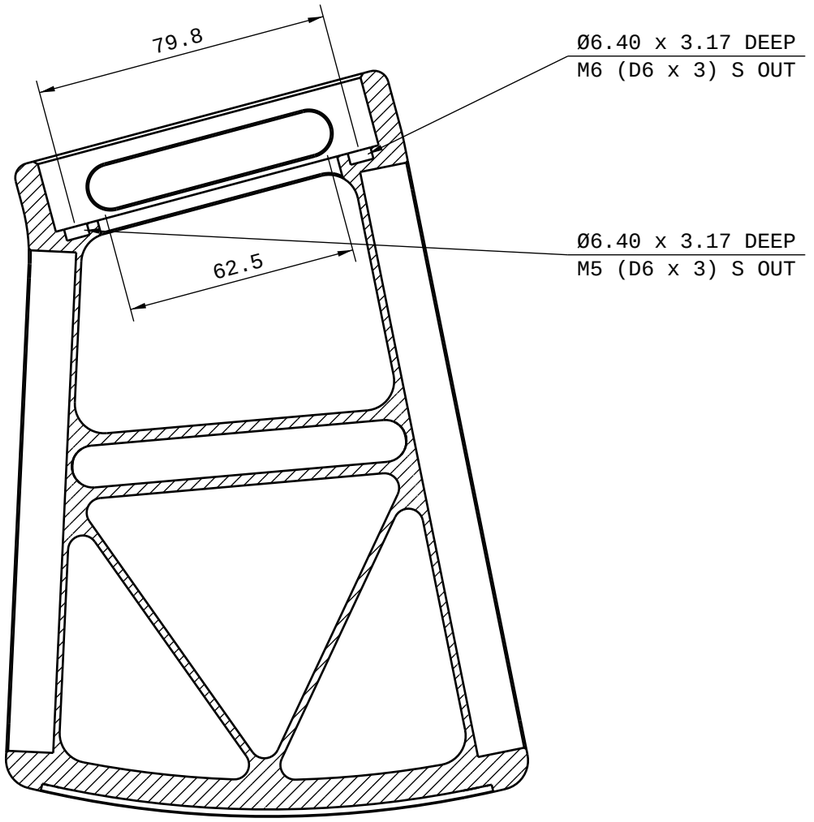
RIGHT

ULTRA LEG, SCALE 3:5. 19.00 PLATE, C-CHANNEL RAILS (2.0 FLANGES), 7.0 TRUSS WEB. SOCKET IS TILTED 15 DEG TO MATCH THE SEAT. NEAR FACE = -Y (FACING VIEWER IN FRONT VIEW). SOCKET LENGTHS AND SECTIONS: SHEET 6.

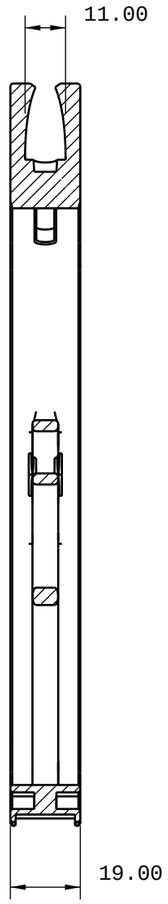
DEPT.	TECHNICAL REFERENCE	CREATED BY	APPROVED BY
CW&T		Che-Wei Wang	10/16/26
		DOCUMENT TYPE: Fabrication drawing - ULTRA LEG	DOCUMENT STATUS: FOR FABRICATION
CW&T		TITLE: Ultra Bench T	DWG NO.: UBT-001
		REV.: A	DATE OF ISSUE: 10/16/26 / SHEET: 5/8



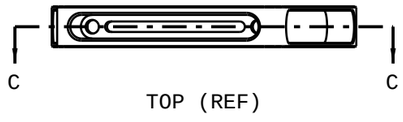
SECTION D-D, SCALE 1:1



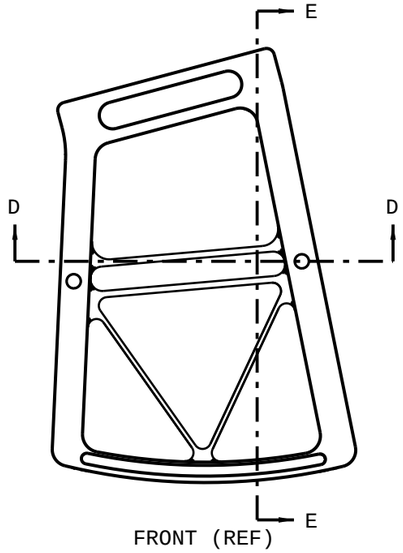
SECTION C-C (Y=0), SCALE 3:5



SECTION E-E, SCALE 3:5

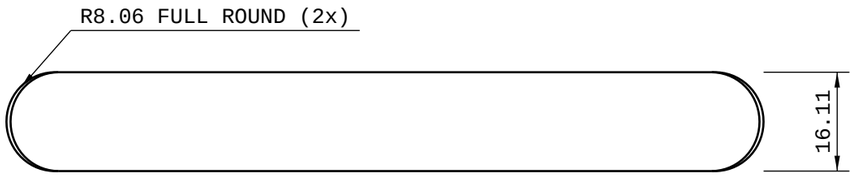


TOP (REF)

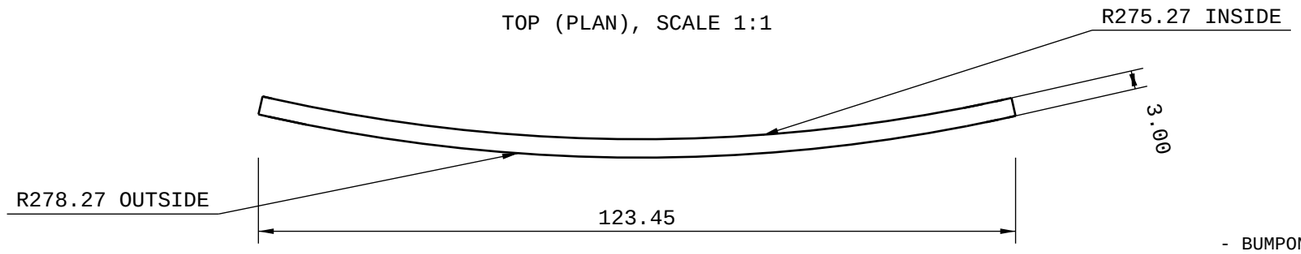


FRONT (REF)

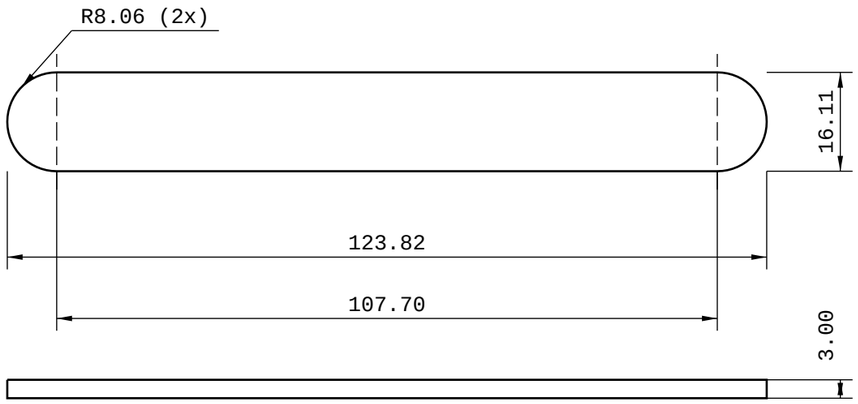
DEPT. CW&T	TECHNICAL REFERENCE	CREATED BY Che-Wei Wang	APPROVED BY 10/16/26
		DOCUMENT TYPE Fabrication drawing - ULTRA LEG SECTIONS	DOCUMENT STATUS FOR FABRICATION
CW&T		TITLE Ultra Bench T	DWG NO. UBT-001
		REV. A	DATE OF ISSUE 10/16/26 / SHEET 6/8



TOP (PLAN), SCALE 1:1



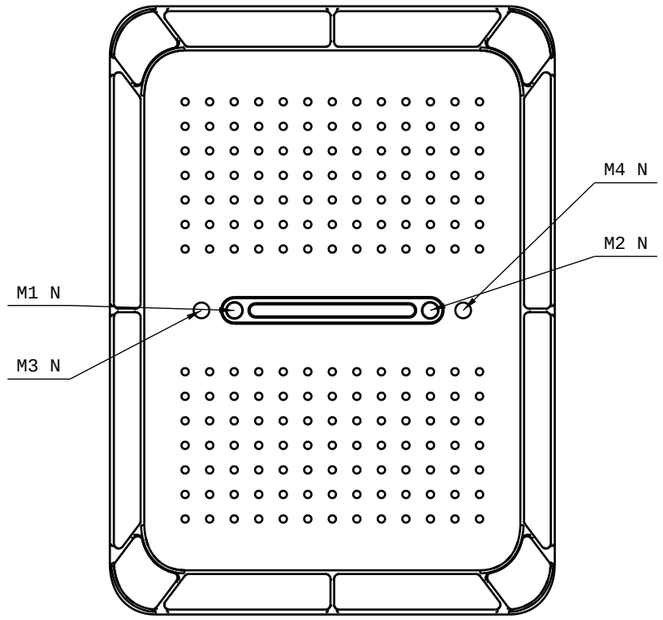
FRONT (AS INSTALLED), SCALE 1:1



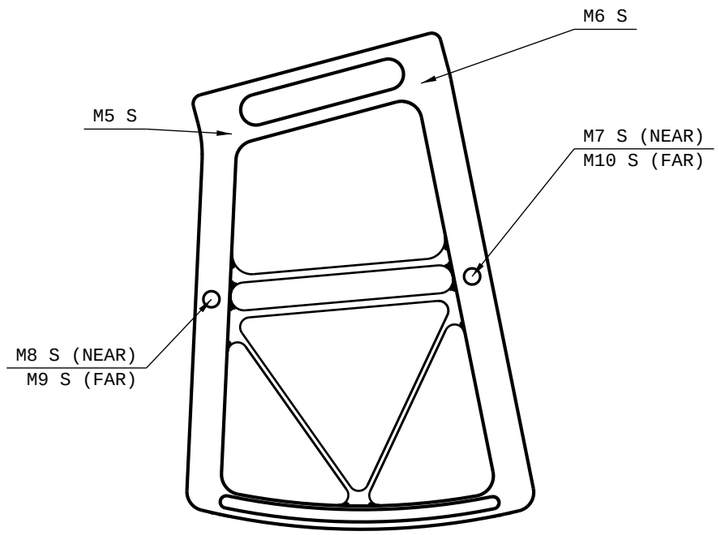
FLAT PATTERN, SCALE 1:1 (EDGE VIEW BELOW)

- BUMPON: ITEM 3, TPR STRIP 3.0 THK. CUT FLAT TO THE PATTERN SHOWN, THEN BEND INTO THE LEG FOOT GROOVE (SHEET 5: R275.27 CEILING, 16.12 WIDE, 2.0 DEEP).
- FLAT LENGTH 123.82 IS THE NEUTRAL-AXIS (MID-THICKNESS) ARC LENGTH OVER 25.63 DEG. INSIDE FACE 123.15, OUTSIDE FACE 124.49.
- INSTALLED, THE STRIP STANDS 1.00 PROUD OF THE LEG SOLE (R277.27) AND IS THE ONLY FLOOR CONTACT.
- ENDS ARE FULL ROUND. BOND INTO THE GROOVE WITH AN ADHESIVE RATED FOR TPR TO ALUMINUM; CLEAN THE GROOVE WITH IPA FIRST, SEAT THE STRIP FULLY AGAINST THE CEILING, WIPE SQUEEZE-OUT. CONFIRM FIT ON FIRST ARTICLE.

DEPT. CW&T	TECHNICAL REFERENCE	CREATED BY Che-Wei Wang	APPROVED BY 10/16/26
		DOCUMENT TYPE Fabrication drawing - BUMPON	DOCUMENT STATUS FOR FABRICATION
CW&T		TITLE Ultra Bench T	DWG NO. UBT-001
		REV. A	DATE OF ISSUE 10/16/26 / SHEET 7/8



TOP, UNDERSIDE (M1-M4)



LEG, FRONT (-Y FACE) (M5-M10)

Magnet Schedule (bores D6.40 +0.05/-0)

ID	Size	Item	Part	Location	Bore dp	Recess	Pole
M1	D6 x 6	ITEM 4	TOP	TENON END FACE	6.34	0.34	N OUT
M2	D6 x 6	ITEM 4	TOP	TENON END FACE	6.34	0.34	N OUT
M3	D6 x 1	ITEM 6	TOP	POCKET CEILING	1.00	0.00	N OUT
M4	D6 x 1	ITEM 6	TOP	POCKET CEILING	1.00	0.00	N OUT
M5	D6 x 3	ITEM 5	LEG	SOCKET FLOOR	3.17	0.17	S OUT
M6	D6 x 3	ITEM 5	LEG	SOCKET FLOOR	3.17	0.17	S OUT
M7	D6 x 1.5	ITEM 7	LEG	-Y (NEAR) FACE	1.64	0.14	S OUT
M8	D6 x 1.5	ITEM 7	LEG	-Y (NEAR) FACE	1.64	0.14	S OUT
M9	D6 x 1.5	ITEM 7	LEG	+Y (FAR) FACE	1.64	0.14	S OUT
M10	D6 x 1.5	ITEM 7	LEG	+Y (FAR) FACE	1.64	0.14	S OUT

- ALL MAGNETS: NdFeB N52, Ni-Cu-Ni PLATED, AXIALLY MAGNETIZED, D6.00.
- 'OUT' = THE FACE AT THE BORE MOUTH, TOWARD THE MATING PART.
- TOP MAGNETS M1-M4: N OUT. LEG MAGNETS M5-M10: S OUT.
- M1/M2 (TOP TENON) MATE WITH M5/M6 (LEG SOCKET) IN THE T CONFIGURATION.
- M3/M4 (TOP CEILING) MATE WITH M7-M10 (LEG FACES) WHEN THE LEG IS FOLDED INTO THE TOP'S POCKET, EITHER FACE UP.
- RECESS = BORE DEPTH - MAGNET THICKNESS. MAGNET SITS BELOW THE MOUTH.
- MARK EACH MAGNET'S OUT FACE BEFORE INSTALLING. VERIFY ATTRACTION WITH THE MATING PART BEFORE BONDING.
- EPOXY MAGNETS IN: SEAT ON BORE FLOOR, BOND WITH TWO-PART EPOXY, WIPE SQUEEZE-OUT FLUSH. NO EPOXY ON THE OUT FACE.

DEPT.	TECHNICAL REFERENCE	CREATED BY	APPROVED BY	
CW&T		Che-Wei Wang	10/16/26	
		DOCUMENT TYPE	DOCUMENT STATUS	
		Fabrication drawing - MAGNETS	FOR FABRICATION	
CW&T		TITLE	DWG NO.	
		Ultra Bench T	UBT-001	
		REV.	DATE OF ISSUE	SHEET
		A	10/16/26	8/8